Trails — Product Requirements Document

v0.2.1 · Real trails, beautifully captured

Status: Implementation specification
Platform: Native iOS
Release type: Product-completion and reliability release
Supersedes: v0.2.0 and earlier v0.2.1 drafts wherever requirements conflict

> Record a journey. Leave moments along the way. Return to what you actually experienced.

1. Release intent

Trails v0.2.1 turns the existing interface and capture capabilities into a product that can be demonstrated using real recordings.

The release must replace hardcoded experiences with a complete loop:

```
Start a trail
→ Capture photos, videos, voice, and notes
→ Record a path when available
→ Optionally scan an indoor space
→ Finish and save
→ Reopen and revisit
```

The release standard is **completeness**: fewer capabilities, implemented well enough to trust.

Video is mandatory. A good camera experience is mandatory. RoomPlan is optional.

A beautiful screen that does not represent saved user content is not a completed feature.

2. Product vision

Trails is a personal journal of places as people experience them.

A trail connects:

- The journey.
- The moments that mattered.
- The person recording them.

- The context available from location or spatial capture.

The long-term vision includes community perspectives and the preservation of personal and cultural memory. This release establishes the prerequisite: recording something genuine and preserving it reliably.

Primary user outcome

> “I can go somewhere, capture what matters, and later show someone how I experienced it.”

Demonstration outcome

The creator can record a real trail on an iPhone, close the app, reopen it, and present the resulting experience without prepared fixtures or developer intervention.

3. Product principles

Real content first

Production screens use actual records, assets, timestamps, and counts.

Capture before organisation

No title, profile, place selection, or scan is required before starting.

Media quality matters

Photos, video, and sound are the experience—not secondary attachments to geometry.

Spatial capture enriches a trail

A trail remains valid without a room model, floor plan, or geographic path.

Preserve independently

Failure of one optional component must not discard successfully captured content.

Honest states

The interface distinguishes recording, saving, processing, saved, interrupted, and failed.

Continuity

Recording, reviewing, editing, and revisiting operate on the same trail record.

4. Scope and release boundary

Required for v0.2.1

- Real local trail library.
- Start, finish, save, and reopen.
- Photo capture.
- Video capture with audio.
- Voice moments.
- Written moments.
- Geographic path recording while the trail is active in the foreground, when available.
- Optional indoor scanning using existing working capabilities.
- Actual media playback.
- Adaptive trail detail.
- Titles, introductions, and cover selection.
- Moment editing and deletion.
- Whole-trail deletion.
- Presentation mode.
- Interrupted-session recovery.
- Permission and storage failure handling.
- On-device persistence.

Secondary, only after the core release gate passes

- Place collections.
- Library search.
- Date filters.
- Geographic collection map.
- Importing existing media.
- Video trimming.
- Advanced path animation.

Hide these features if unfinished.

Out of scope

- Public publishing and internet sharing links.
- Logos integration.
- Account synchronisation.

- AI narration, films, or generated heritage descriptions.
- Turn-by-turn navigation.
- Automatic scan merging.
- Automatic alignment between independent recordings.
- Followers, ratings, and contribution scores.
- Guaranteed background recording.
- Advanced professional camera modes.
- Permanent archival guarantees.

5. Core product objects

Trail

One recording session containing:

- Start and end times.
- Creator reference.
- Optional title and introduction.
- Moments.
- Optional geographic samples.
- Optional spatial sessions.
- Lifecycle state.

A trail can contain moments without a path, or a path without moments.

Moment

A meaningful point or interval within a trail.

May contain:

- Photo.
- Video.
- Voice recording.
- Text.
- Optional positioning.

Media recorded during a session creates moments automatically. The user should not need to create an empty moment before pressing Record.

Spatial session

An optional indoor capture associated with a trail.

Contains whichever outputs were actually produced:

- Geometry.
- Local path.
- Plan.
- Coordinate-frame information.

Place

An optional grouping for related trails.

Sharing a place does not mean sharing geometry or coordinates.

6. Information architecture

```
My Trails
  ├── Saved trails
  ├── Recoverable drafts
  ├── Start a trail
  └── Trail detail
       ├── Moments
       ├── Available map/spatial views
       ├── Edit
       └── Present

Active Trail
  ├── Photo
  ├── Video
  ├── Voice
  ├── Note
  ├── Optional indoor scan
  ├── Captured moments
  └── Finish

Settings
  ├── Permissions
  ├── Storage information
  ├── Privacy
  └── App information
```

Do not add empty destinations solely to make the app appear larger.

7. First launch and empty state

Initial screen

Suggested copy:

> “Keep the places you experience, and the moments you want to remember.”

Primary action:

> Start a trail

Requirements

- No mandatory account.
- No mandatory profile.
- No upfront permission wall.
- No hardcoded Haveli trail in the personal library.
- No simulated activity or counts.
- Optional examples must be explicitly labelled and separate from personal records.

A display name may be added later. It must not block capture.

8. Starting a trail

On Start:

1. Create a stable trail ID.
2. Persist a draft.
3. Record the start timestamp.
4. Request only the permissions needed for the next action.
5. Begin available foreground location collection.
6. Open the capture experience.

Location denial does not block photography, video, voice, or notes.

Camera denial does not block text or other permitted inputs.

Default title

Use a neutral title derived from the real recording date and time.

Do not infer a place name unless supported by actual data or supplied by the user.

9. Active recording experience

Default layout

```
Trail active                                Finish
00:04:12

                  Camera preview

               [Location status]

[Latest saved moment]                [All moments]

           PHOTO       VIDEO       VOICE

                     [Capture]

                  Add a note

               Scan indoor space
```

Exact visual treatment may follow the existing design language. Interaction behaviour must remain clear.

Persistent information

- Trail elapsed time.
- Whether video or voice recording is active.
- Relevant media recording duration.
- Moment count.
- Location status.
- Saving/error feedback.

Two separate recording lifecycles

- The trail is the overall journey.
- A video or voice recording is one moment within it.

Stopping a clip does not finish the trail.

Finishing a trail must safely stop and save any active media recording.

10. Camera experience

10.1 Preview and controls

Required:

- Responsive live preview.
- Correct orientation.
- Predictable framing.
- Clear Photo/Video mode selection.
- Front/rear camera selection before recording.
- Supported zoom controls.
- Automatic focus and exposure.
- Tap-to-focus/expose where supported.
- Flash/torch controls only where available.
- Distinct idle, recording, and finalising states.

Do not expose unavailable controls as decorative buttons.

10.2 Capture implementation direction

Use a native camera pipeline, with Apple's AVCam approach as the reference established in the earlier camera amendment.

Keep camera management separate from trail persistence.

Do not use screen recording or a screenshot of the preview as the photo/video implementation.

10.3 Mode switching

- Preserve the active trail.
- Do not lose saved moments.
- Disable incompatible switches during an active recording.
- Explain why an action is unavailable when necessary.
- Do not unexpectedly stop recording because the user tapped another mode.

11. Photo requirements

Capture flow

```
Take photo
→ Persist image
→ Create moment
→ Generate thumbnail
→ Confirm saved
→ Continue trail
```

Requirements:

- Save the actual captured image.
- Preserve correct orientation.
- Display an actual thumbnail.
- Allow immediate review.
- Allow deletion.
- Allow an optional caption afterward.
- Keep the user inside the same trail.

A camera failure must not create a seemingly valid photo moment backed by a missing file.

12. Video requirements

Video is a release-blocking capability.

12.1 Recording

Required:

- Dedicated Video mode.
- Obvious start/stop control.
- Visible recording indicator.
- Accurate elapsed recording time.
- Audio recording when authorised.
- Predictable handling of orientation.
- Clear finalising state after Stop.
- Ability to record several clips in one trail.

12.2 Quality

Initial product target:

- 1080p at 30 fps where supported and validated.
- A supported fallback on the declared device matrix.

- Original captured files retained.
- Separate lightweight thumbnails for browsing.

The supported device matrix must be defined before release.

12.3 Duration

- No arbitrary 30-second cap.
- No promise of unlimited recording.
- Communicate any implemented limit before recording.
- Test sustained recording on supported devices.
- Handle storage exhaustion explicitly.

12.4 Camera switching

Front/rear selection is required before recording.

Switching cameras during a clip is not required for v0.2.1. Disable it during recording rather than risking a broken clip.

12.5 Audio permission denied

Allow video without audio after a clear explanation:

> “Microphone access is off. This video will be recorded without sound.”

Do not silently imply that sound was captured.

12.6 Finalisation

After Stop:

1. Finalise the file.
2. Store it in managed trail storage.
3. Persist metadata and attachment references.
4. Generate its preview.
5. Confirm the moment is saved.

Do not report success merely because recording stopped.

13. Voice and text moments

Voice

- Explicit start/stop.
- Visible duration.
- Saved audio file.
- Working playback.
- Optional title or note.
- No generated transcript unless explicitly implemented later.
- No waveform pretending to represent unanalysed audio.

Video and standalone voice recording must not compete for the microphone. Block incompatible actions clearly.

Text

- Quick-entry sheet.
- Save exact user text.
- Edit and delete.
- Optional label.
- No automatic invented descriptions.

Writing a note must not end the trail.

14. Moment positioning and ordering

Position rules

Attach only available, valid context:

- Geographic coordinate.
- Location timestamp and accuracy information.
- Local spatial position.
- Coordinate-frame reference.
- Capture timestamp.

If no suitable position exists, save the moment without one.

Do not:

- Reuse a stale coordinate without qualification.
- Place all moments at the trail origin.
- Fabricate indoor alignment.

Time-based media

Video and voice have start and end times.

Store available positioning samples for their interval. Do not pretend a moving clip represents one exact position throughout.

Ordering

Default to recording order.

If presentation reordering is implemented:

- Preserve original timestamps and capture order.
- Store presentation order separately.

15. Geographic path

Required behaviour

- Record available location samples during an active foreground trail.
- Persist samples incrementally.
- Preserve timestamps and accuracy metadata.
- Show location unavailable/acquiring states.
- Keep recording moments when location is unavailable.

Rendering

- Render actual recorded samples.
- Mark or break meaningful gaps.
- Do not smooth a path into a route the user did not record.
- Do not connect interrupted segments in a misleading way.

A geographic path is a record, not a navigation instruction.

16. Optional indoor scanning

Entry

“Scan indoor space” opens a dedicated subflow attached to the current trail.

Requirements

- Keep the parent trail saved.
- Preserve moments already captured.
- Explicitly coordinate camera ownership.
- Return to the same trail after scanning.
- Attach only assets that were actually generated.
- Track independent local coordinate frames.

Camera transitions

Normal media capture and spatial scanning must not independently attempt to own the camera.

A shared capture coordinator manages:

- Starting.
- Stopping.
- Releasing resources.
- Entering scan mode.
- Returning to media mode.
- Handling failures.

Mark local tracking discontinuities. Do not imply continuous indoor tracking across an unsupported transition.

Scan failure

A failed scan does not invalidate photos, videos, voice, notes, or geographic samples.

17. Captured-moments tray

During a trail, the user can inspect saved moments.

Show:

- Actual thumbnail or media-type indicator.
- Capture time.
- Optional label.
- Save/error state.

Actions:

- Preview.
- Add/edit caption.

- Delete with appropriate confirmation.
- Return to recording.

Prevent incompatible playback or capture actions during active media recording. Never silently terminate a recording to open a preview.

18. Finish and save

Finish sequence

```
Finish
→ Stop active media safely
→ Persist pending data
→ Finalise available assets
→ Update lifecycle state
→ Open review
```

Feedback

Use specific language:

> “Trail saved on this iPhone.”

For partial completion:

> “Your moments are saved. The indoor scan is still processing.”

For recoverable attachment failure:

> “Trail saved. One recording needs attention.”

Empty sessions

If there are no usable moments, path samples, or spatial assets:

- Explain that nothing was captured.
- Offer Continue or Discard.

Do not populate an empty trail with sample content.

19. Saved trail detail

Always available

- Actual title.
- Recording date.
- Optional introduction.

- Actual moments.
- Edit.
- Delete.
- Presentation mode.

Adaptive views

Saved content	Available experience
Photos, video, voice, notes	Moment sequence
Geographic path	Route map
Indoor geometry	3D viewer
Plan output	Plan viewer
Local path	Compatible local spatial view
Missing optional asset	Clear unavailable state; other content still opens

Do not show empty 3D, Plan, or Map tabs to suggest nonexistent content.

Editing

Required:

- Rename.
- Edit introduction.
- Choose cover from actual content.
- Edit moment text.
- Delete moments.
- Delete trail.

Trimming, imported attachments, and complex restructuring are not release requirements.

20. Revisit and presentation

Moments view

The dependable default:

- Browse chronologically.
- Open photos.
- Play video and voice.
- Read notes.
- See associated context when available.

Video playback

Required:

- Play/pause.
- Seek/scrub.
- Duration and progress.
- Full-screen playback.
- Correct orientation.
- Return to the same moment.

Presentation mode

- Hide editing controls.
- Retain obvious navigation.
- Keep playback controls accessible.
- Avoid unexpected audio autoplay.
- Exit clearly.
- Preserve current position on exit.

No AI montage or cinematic replay is required.

The user's captured material should carry the experience.

21. Library

Each item derives from a persisted trail:

- Title.
- Date.
- Actual cover or neutral placeholder.
- Actual moment count.
- Optional place.
- Lifecycle state.

Required:

- Open the correct record.
- Show multiple independent trails.
- Distinguish recoverable drafts.
- Rename.
- Delete.

Do not load full-resolution videos or spatial assets just to render library cards.

22. Data and persistence

Minimum records

```
Trail
  id
  schemaVersion
  creatorId?
  title
  introduction?
  startedAt
  endedAt?
  updatedAt
  state
  placeId?
  momentIds[]
  geographicTrackReference?
  spatialSessionReferences[]
  coverReference?
```

```
Moment
  id
  trailId
  type
  startedAt
  endedAt?
  label?
  text?
  mediaReferences[]
  geographicContext?
  localSpatialContext?
  originalOrder
  presentationOrder?
```

```
MediaAsset
  id
  type
  relativeFileReference
  thumbnailReference?
  duration?
  dimensions?
  orientation?
  persistenceState
  processingState
```

Persistence rules

- Create the draft before capture.
- Save moments incrementally.
- Persist path samples incrementally.
- Use atomic metadata updates.
- Separate originals from derived previews.
- Version the storage format.
- Resolve relative asset references consistently.
- Recover incomplete processing where possible.
- Avoid deleting assets still referenced elsewhere.
- Keep storage cleanup consistent with metadata deletion.

23. Lifecycle and recovery

```
Draft
→ Recording
→ Finalising
→ Saved
```

Additional states:

- Interrupted.
- Processing optional assets.
- Saved with recoverable errors.
- Unrecoverable failure.

Interruption policy

For app backgrounding, calls, camera loss, or other interruptions:

- Preserve completed content.
- Stop/finalise active media where possible.
- Record the interruption.
- Explain what happened on return.
- Let the user continue the trail or finish it.

v0.2.1 does not promise uninterrupted background capture.

After relaunch, unfinished sessions must not disappear or masquerade as fully completed recordings.

24. Privacy and permissions

- No required account.
- No automatic uploads.
- Store media in managed local app storage.
- Request permissions contextually.
- Explain denied permissions without blocking unrelated capabilities.
- Document actual backup behaviour.
- Explain the consequences of deleting the app.
- Do not claim permanent preservation or unsupported encryption guarantees.

Deleting a trail requires confirmation and removes its owned local content.

25. UX quality requirements

- Preserve the existing visual language where it works.
- Make recording controls unmistakable.
- Never use colour alone to communicate state.
- Support readable text sizing.
- Provide accessible control labels.
- Use sufficiently large touch targets.
- Respect reduced-motion preferences.
- Avoid transitions that delay capture.
- Never conceal a failed save behind a celebratory animation.

Performance acceptance

Test on the declared supported devices for:

- Responsive preview.
- Reliable repeated capture.
- Sustained video recording.
- Memory stability.
- Thumbnail loading.
- Spatial/media transitions.
- Low-storage behaviour.

Measured results, not optimistic assumptions, determine supported configurations.

26. Production-content audit

Before adding polish, inspect every screen.

Existing element	Required treatment
Hardcoded Haveli title/story	Remove from personal-data flows
Sample photos/video	Keep only in explicitly labelled examples
Fake moment counts	Replace with persisted counts
Simulated audio duration/waveform	Replace with actual metadata or remove
Dummy map coordinates	Remove
Fake save/share/publish confirmation	Remove
Static room preview	Show only as a labelled example
Unsupported controls	Implement or hide

Development fixtures may remain in previews and test tooling. Production must not silently fall back to them.

27. Implementation sequence

Stage 1 — Establish real records

- Audit hardcoded content.
- Create draft and saved trail lifecycle.
- Make the library open actual trail IDs.
- Implement incremental persistence and recovery.

Stage 2 — Complete the camera

- Photo capture.
- Video with audio.
- Camera switching, focus/exposure, zoom.
- Recording/finalising states.
- Durable storage and playback.

Stage 3 — Complete the journey

- Voice and text moments.
- Geographic samples.
- Finish/review.
- Editing and deletion.

- Presentation mode.

Stage 4 — Reconnect spatial capabilities

- Optional scan subflow.
- Shared camera coordination.
- Conditional 3D/Plan.
- Independent scan failure handling.

Stage 5 — Validate on real trails

- Record outside the development environment.
- Restart and replay.
- Test permissions, interruptions, and storage.
- Fix failures before adding secondary discovery features.

28. Release acceptance tests

A. Complete real trail

1. Start a trail.
2. Take two photos.
3. Record a video with speech.
4. Move and record another clip.
5. Add voice and text.
6. Finish.
7. Force-close the app.
8. Reopen without internet.
9. Play every item.

Expected: correct content, order, sound, orientation, and metadata.

B. No RoomPlan

Complete Test A without scanning.

Expected: no degraded or blocked core experience.

C. No location

Deny location and record media.

Expected: valid trail without fabricated coordinates.

D. Silent video

Deny microphone permission.

Expected: explicit warning, usable silent video, no false audio indication.

E. Optional scan

Capture media, scan an indoor space, then capture more media.

Expected: one trail with preserved moments and correctly associated spatial assets.

F. Scan failure

Interrupt spatial capture.

Expected: other content remains saved and reviewable.

G. Interruption recovery

Interrupt an active recording and relaunch.

Expected: completed content survives; incomplete content is identified honestly.

H. Multiple independent trails

Record two trails and reopen both after restart.

Expected: each loads its own content, not a shared static detail screen.

I. Storage failure

Test insufficient storage during capture/finalisation.

Expected: clear failure, no false success, no deletion of unrelated saved content.

J. Fresh installation

Expected: honest empty state, working Start action, no fabricated personal history.

29. Definition of done

Trails v0.2.1 is ready when someone can:

> Go somewhere, record a real experience with a good camera, leave meaningful moments, save the journey, and confidently show it again later.

The demonstration must use a newly recorded trail—not a prepared Haveli fixture.

The camera captures the experience. The moments give it meaning. Location and spatial scanning add context. None of those optional layers should prevent the real product from working.